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EXPERIMENT 5

Aim: Implement least privilege principle to minimize the number of permissions granted to users by using database tool.

Theory:

Implementing the Principle of Least Privilege (PoLP) in a Database Using MySQL Workbench

Understanding the Principle of Least Privilege (PoLP)

The Principle of Least Privilege (PoLP) is a fundamental security approach that restricts users, applications, and systems to only the essential permissions required for their specific functions. In database management, implementing PoLP reduces exposure to security threats, prevents unauthorized modifications, and enhances data integrity.

By leveraging MySQL Workbench, administrators can systematically assign and control access rights, ensuring that users operate within predefined boundaries to maintain a secure and efficient database environment.

Stage 1: Identify User Roles and Permissions

1. Analyze job functions and determine the minimum permissions each role needs.
2. Create a list of database operations each role requires (e.g., READ, WRITE, DELETE).

Stage 2: Create Users and Assign Minimal Permissions

Use a database management tool such as **MySQL Workbench**, **SQL Server Management Studio (SSMS)**, or **pgAdmin** to create and configure users with only necessary privileges.

A. Example 1: MySQL

-- Create a new user

```
CREATE USER 'app_user'@'localhost' IDENTIFIED BY 'StrongPassword';
```

-- Grant only SELECT permission on a specific database

```
GRANT SELECT ON company_db.* TO 'app_user'@'localhost';
```

-- Apply changes

```
FLUSH PRIVILEGES;
```



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Stage 3: Implement Role-Based Access Control (RBAC)

Instead of assigning permissions directly to users, assign them to roles.

Example: Assigning Roles in MySQL

```
-- Create a role with minimal privileges
CREATE ROLE read_role;

-- Grant SELECT permission to the role
GRANT SELECT ON company_db.* TO read_role;

-- Assign the role to a user
GRANT read_role TO 'app_user'@'localhost';

-- Set default role (optional)
SET DEFAULT ROLE read_role FOR 'app_user'@'localhost';
```

Stage 4: Regularly Audit and Revoke Unnecessary Permissions

Check User Permissions

□ MySQL:

```
SHOW GRANTS FOR 'app_user'@'localhost';
```

Revoke Excessive Permissions

```
REVOKE INSERT, UPDATE, DELETE ON company_db.* FROM 'app_user'@'localhost';
```

Stage 5: Monitor and Enforce Access Controls

- Use **logging and auditing tools** to track access attempts.
- Enable **Database Activity Monitoring (DAM)**. □
Set up **alerts** for privilege escalations.

Output:



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34 • `select *from students;`

34 • `select *from students;`

Result Grid				
		Filter Rows:	Edit:	
roll_no	name	address	blood_group	
1	Aarav Mehta	Mumbai, India	A+	
2	Sneha Sharma	Delhi, India	B+	
3	Rohan Patel	Ahmedabad, India	O+	
4	Meera Iyer	Chennai, India	AB+	
5	Vikram Singh	Jaipur, India	A-	
6	Pooja Nair	Kolkata, India	B-	
7	Amit Das	Bangalore, India	O-	
8	Ritu Verma	Pune, India	AB-	
9	Kunal Thakur	Hyderabad, India	A+	
10	Neha Joshi	Lucknow, India	B+	
11	Rahul Kapoor	Indore, India	O+	
12	Simran Malhotra	Chandigarh, India	AB+	
13	Arjun Desai	Surat, India	A-	
14	Priya Saxena	Bhopal, India	B-	
15	Manoj Gupta	Nagpur, India	O-	
16	Swati Kulkarni	Patna, India	AB-	
17	Deepak Reddy	Vishakhapatnam, ...	A+	
18	Nisha Yadav	Coimbatore, India	B+	
19	Rakesh Sen	Guwahati, India	O+	
20	Sonia Dutta	Raipur, India	AB+	

36 • `CREATE USER 'root'@'3306' IDENTIFIED BY 'StrongPassword123';`

8 11:14:03 CREATE USER 'root'@'3306' IDENTIFIED BY 'StrongPassword123'

0 row(s) affected

38 • `GRANT SELECT ON student_db.* TO 'root'@'3306';`

9 11:14:40 GRANT SELECT ON student_db.* TO 'root'@'3306'

0 row(s) affected

40 • `SHOW GRANTS FOR 'root'@'3306';`

Result Grid	
Filter Rows:	
Export:	
Grants for root@3306	
▶	GRANT USAGE ON *.* TO 'root'@'3306'
	GRANT SELECT ON `student_db`.* TO 'root'...



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```
• SELECT * FROM student_db.students;
• SELECT * FROM student_db.students;
47 • INSERT INTO students VALUES (21, 'Raj Malhotra', 'Bangalore, India', 'O+');
47 • INSERT INTO students VALUES (21, 'Raj Malhotra', 'Bangalore, India', 'O+');
-- User 2: data_entry (Insert & Select Privileges);
CREATE USER 'data_entry'@'localhost' IDENTIFIED BY 'EntryPass456';
16 11:34:05 CREATE USER 'data_entry'@'localhost' IDENTIFIED BY 'EntryPass456' 0 row(s) affected
GRANT SELECT, INSERT ON student_db.* TO 'data_entry'@'localhost';
• FLUSH PRIVILEGES;
18 11:39:06 FLUSH PRIVILEGES
-- User 3: admin_user (Full Control)
CREATE USER 'admin_user'@'3306' IDENTIFIED BY 'AdminPass789';
19 11:43:30 CREATE USER 'admin_user'@'3306' IDENTIFIED BY 'AdminPass789' 0 row(s) affected
GRANT ALL PRIVILEGES ON student_db.* TO 'admin_user'@'3306';
22 11:45:52 GRANT ALL PRIVILEGES ON student_db.* TO 'admin_user'@'3306'
-- Verify Permissions for Each User
• SHOW GRANTS FOR 'root'@'3306';
• SHOW GRANTS FOR 'data_entry'@'3306';
• SHOW GRANTS FOR 'admin_user'@'3306';
```

Result Grid | Filter Rows: | Export:

Grants for root@3306
▶ GRANT USAGE ON *,* TO 'root'@'3306'
GRANT SELECT ON 'student_db',* TO 'root'...

Result Grid | Filter Rows: | Export:

Grants for data_entry@3306
▶ GRANT USAGE ON *,* TO 'data_entry'@'3306'

Result Grid | Filter Rows: | Export:

Grants for admin_user@3306
▶ GRANT USAGE ON *,* TO 'admin_user'@'3306'
GRANT ALL PRIVILEGES ON 'student_db',* TO...



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```
-- Role-Based Access Control (RBAC)

-- Create Roles

CREATE ROLE read_role;
CREATE ROLE entry_role;
CREATE ROLE admin_role;

40 11:53:28 CREATE ROLE read_role
41 11:53:31 CREATE ROLE entry_role
42 11:53:35 CREATE ROLE admin_role
-- Assign Privileges to Roles

GRANT SELECT ON student_db.* TO read_role;
GRANT SELECT, INSERT ON student_db.* TO entry_role;
GRANT ALL PRIVILEGES ON student_db.* TO admin_role;

43 11:58:18 GRANT SELECT ON student_db.* TO read_role
44 11:58:20 GRANT SELECT, INSERT ON student_db.* TO entry_role
45 11:58:22 GRANT ALL PRIVILEGES ON student_db.* TO admin_role
-- Assign Roles to Users

GRANT read_role TO 'root'@'3306';
GRANT entry_role TO 'data_entry'@'3306';
GRANT admin_role TO 'admin_user'@'3306';

46 12:00:31 GRANT read_role TO 'root'@'3306'
47 12:00:33 GRANT entry_role TO 'data_entry'@'3306'
48 12:00:35 GRANT admin_role TO 'admin_user'@'3306'

-- Set Default Roles

SET DEFAULT ROLE read_role TO 'root'@'3306';
SET DEFAULT ROLE entry_role TO 'data_entry'@'3306';
SET DEFAULT ROLE admin_role TO 'admin_user'@'3306';

50 12:03:40 SET DEFAULT ROLE read_role TO 'root'@'3306' 0 row(s) affected
51 12:03:43 SET DEFAULT ROLE entry_role TO 'data_entry'@'3306' 0 row(s) affected
52 12:03:45 SET DEFAULT ROLE admin_role TO 'admin_user'@'3306' 0 row(s) affected

-- Auditing and Security Checks

SHOW GRANTS FOR 'root'@'3306';
SHOW GRANTS FOR 'data_entry'@'3306';
SHOW GRANTS FOR 'admin_user'@'3306';
```



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Result Grid	Filter Rows:	Export:
Grants for root@3306		
▶	GRANT USAGE ON *.* TO 'root'@'3306'	
	GRANT SELECT ON 'student_db',* TO 'root'...	
	GRANT 'read_role'@'%' TO 'root'@'3306'	

Result Grid	Filter Rows:	Export:
Grants for data_entry@3306		
▶	GRANT USAGE ON *.* TO 'data_entry'@'3306'	
	GRANT 'entry_role'@'%' TO 'data_entry'@...	

Result Grid	Filter Rows:	Export:
Grants for admin_user@3306		
▶	GRANT USAGE ON *.* TO 'admin_user'@'3306'	
	GRANT ALL PRIVILEGES ON 'student_db',* TO...	
	GRANT 'admin_role'@'%' TO 'admin_user'...	

-- To check who has access to what:

```
SELECT user, host FROM mysql.user;
```

Result Grid	Filter Rows:	Export:	Wrap
	user	host	
▶	admin_role	%	
	entry_role	%	
	read_role	%	
	admin_user	3306	
	data_entry	3306	
	root	3306	
	data_entry	localhost	
	mysql.infoschema	localhost	
	mysql.session	localhost	
	mysql.sys	localhost	
	root	localhost	

user 14 x
89 • SHOW VARIABLES LIKE 'general_log';

Result Grid	Filter Rows:	Export:	Wr
	Variable_name	Value	
▶	general_log	ON	



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```
-- monitor login attempts |
```

```
SELECT * FROM mysql.general_log WHERE command_type = 'Connect';
```

Result Grid						
Filter Rows:		Export:		Wrap Cell Content:		
	event_time	user_host	thread_id	server_id	command_type	argument
▶	2025-03-11 12:16:30.554331	[root] @ localhost [::1]	19	1	Connect	BLOB
	2025-03-11 12:16:30.557742	root[root] @ localhost [::1]	19	1	Connect	BLOB
	2025-03-11 12:16:30.736487	[root] @ localhost [::1]	20	1	Connect	BLOB
	2025-03-11 12:16:30.737267	root[root] @ localhost [::1]	20	1	Connect	BLOB
	2025-03-11 12:16:41.231285	[root] @ localhost [::1]	21	1	Connect	BLOB
	2025-03-11 12:16:41.232722	root[root] @ localhost [::1]	21	1	Connect	BLOB
	2025-03-11 12:16:46.246490	[root] @ localhost [::1]	22	1	Connect	BLOB
	2025-03-11 12:16:46.316651	[root] @ localhost [::1]	23	1	Connect	BLOB

```
-- Revoke Unnecessary Permissions
```

```
REVOKE INSERT, UPDATE, DELETE ON student_db.* FROM 'root'@'3306';
```

9 12:28:34 REVOKE INSERT, UPDATE, DELETE ON student_db.* FROM 'root'@'3306'

0 row(s) affected



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```
23 • SELECT *
24 FROM information_schema.schema_privileges
25 WHERE grantee = "'admin_user'@'3306'"
26 AND table_schema = 'student_db';
```

Result Grid					
		Filter Rows:		Export:	Wrap Cell Content: IA
	GRANTEE	TABLE_CATALOG	TABLE_SCHEMA	PRIVILEGE_TYPE	IS_GRANTABLE
▶	'admin_user'@'3306'	def	student_db	SELECT	NO
	'admin_user'@'3306'	def	student_db	CREATE	NO
	'admin_user'@'3306'	def	student_db	DROP	NO
	'admin_user'@'3306'	def	student_db	REFERENCES	NO
	'admin_user'@'3306'	def	student_db	INDEX	NO
	'admin_user'@'3306'	def	student_db	ALTER	NO
	'admin_user'@'3306'	def	student_db	CREATE TEMPORARY TABLES	NO
	'admin_user'@'3306'	def	student_db	LOCK TABLES	NO
	'admin_user'@'3306'	def	student_db	EXECUTE	NO
	'admin_user'@'3306'	def	student_db	CREATE VIEW	NO
	'admin_user'@'3306'	def	student_db	SHOW VIEW	NO
	'admin_user'@'3306'	def	student_db	CREATE ROUTINE	NO
	'admin_user'@'3306'	def	student_db	ALTER ROUTINE	NO

```
23 • SELECT *
24 FROM information_schema.schema_privileges
25 WHERE grantee = "'root'@'3306'"
26 AND table_schema = 'student_db';
```

Result Grid					
		Filter Rows:		Export:	Wrap Cell Content: IA
	GRANTEE	TABLE_CATALOG	TABLE_SCHEMA	PRIVILEGE_TYPE	IS_GRANTABLE
▶	'root'@'3306'	def	student_db	SELECT	NO

Conclusion:

Thus we implemented PoLP (Principle of least Privilege) in MySQL Workbench. Aim: Implement least privilege principle to minimize the number of permissions granted to users by using database tool. give conclusion for this above aim.